

HW Class 6 (R Functions)

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library(bio3d)
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# install.packages("bio3d")
library(bio3d)
# INPUTS: The PDB ID assigned to the proteins
# WHAT THIS FUNCTION DOES AND HOW TO USE IT: Reads each PDB structure, trims it to chain A and
# OUTPUT: Gives a plot of the B-factor of a protein along the chain A using the CA atoms.
protein_analysis <- function(pdb_id) {
  # Access the protein structure
  pdb <- read.pdb(pdb_id)

  # Trim the protein to Chain A and CA atoms
  pdb_chain <- trim.pdb(pdb, chain = "A", eley = "CA")

  # Extract B-factor
  pdb.b <- pdb_chain$atom$b

  # Plot B-factor
  plotb3(pdb.b, sse = pdb_chain, typ = "l", ylab = "Bfactor")
}
```

```
protein_analysis("1AKE")
```

Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

